TCAD Simulations of Radiation Damage in 4H-SiC

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2024 Austrochip Workshop on Microelectronics (Austrochip) | 979-8-3315-1617-8/24/$31.00 ©2024 IEEE | DOI: 10.1109/AUSTROCHIP62761.2024.10716221

***Abstract*—In this paper we present simulation based radiation damage modeling of 4H silicon carbide (SiC) using the technology computer aided design (TCAD) tools for up to 1 kV forward and backward bias. After verifying the TCAD framework from Global TCAD Solutions (GTS) against Sentaurus simulations for silicon we use it to approximate measurements of neutron-irradiated 4H-SiC particle detectors, i.e., p-i-n diodes. Based on our simulations we are not only able to evaluate the accuracy of the predictions but also to provide an explanation for the almost negligible current of radiated devices under high forward bias.**

***Index Terms*—TCAD simulations, 4H silicon carbide (SiC), particle detector, radiation damage**

I. Introduction

Particle accelerators are a crucial component to explore the limits of the standard model of physics. In order to develop proper extensions, scientists use information that are gathered from very rare particle collision events. To improve the scientific output of the experiments the luminosity of future colliders, like the Future Circular Collider (FCC) at CERN, will be increased significantly. Such high particle fluxes, which are nowadays already encountered in medical accelerators, lead, however, to more radiation damage in the detectors. In order to avoid frequent replacements and a performance loss during operation, materials that feature an improved radiation hardness and can be used to build particle detectors are highly desired. One very promising candidate is silicon carbide (SiC). Due to its material properties, such as a wide bandgap (low leakage) and high charge carrier mobility, devices outperforming the existing silicon ones are enabled.

The key question in using SiC, or more specifically, the most commonly used polytype 4H-SiC, for high energy physics experiments is in which degree the performance degrades while being exposed to ionizing radiation. Recent measurements [1] revealed that at fluences higher than $10^{15}\,\mathrm{n_{eq}/cm^2}$ the capacitance of a diode is absolutely constant for a bias of $-1\,\mathrm{kV}$ to $1\,\mathrm{kV}$ (see Fig. 2a). Furthermore, it is possible to heavily forward bias the device without seeing significant conduction (see Fig. 3a). In order to understand these observations, technology computer aided design (TCAD) simulations can be beneficial. In the past the latter have been used to describe the degradation of Schottky diodes [2], [3] or neutron detectors but only by a single defect [4]. Only recently, a more elaborate model has been proposed to describe the radiation damage in p-i-n diodes [5], however, no comprehensive simulations have been conducted.

In this paper, we will, thus, present an evaluation of radiation damage in 4H-SiC under high forward and reverse biases by means of TCAD simulations. At first we verify the accuracy of the simulation framework from Global TCAD Solutions [6] by comparing the achieved results for a silicon sensor to established simulation tools. Afterwards, we focus on recreating measurements of 4H-SiC. Qualitatively, good matches are achieved and the simulation results allow us to hypothesize possible causes for the observed phenomena. In this publication, we first provide some background information in Section II before showing the achieved simulation results in Section III and concluding the paper in Section IV.

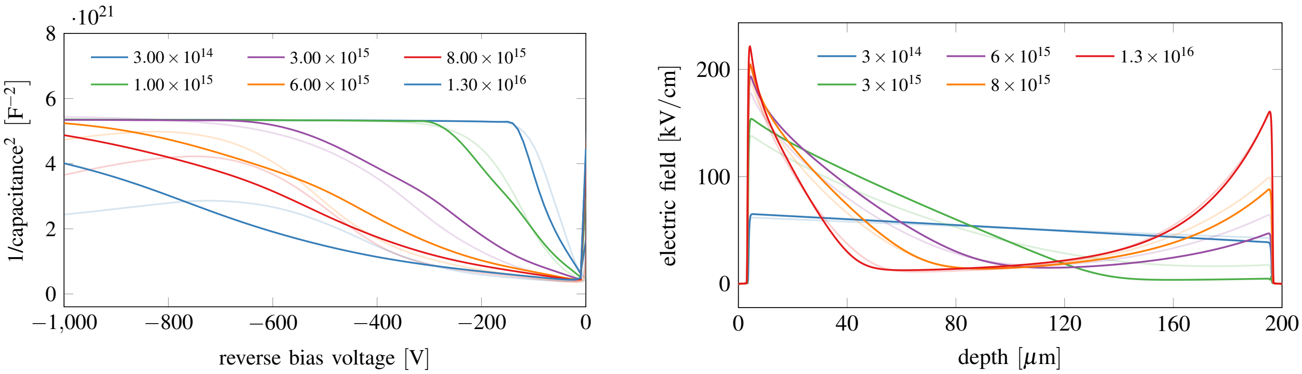
II. Background

Silicon carbide (SiC) is a wide bandgap material that is heavily deployed in power electronics. It can crystallize in different crystal arrangements, called polytypes, whereas 4H is the most commonly used in industry (bandgap energy $E_g = 3.285\,\mathrm{eV}$ [7]). Due to its high charge carrier mobility and thermal conductivity, it is a very good alternative to the currently established silicon.

The most basic particle detectors are reverse biased p-i-n diodes. Charge carrier pairs generated by the impinging particle in the space charge region are, due to the high electric field, immediately drawn towards the cathode (electrons) resp. anode (holes) and induce a current signal during their drift. For detection purposes the wide bandgap of 4H-SiC is not solely advantageous as fewer charge carriers are generated per unit distance. In the case of a minimum ionizing particle (MIP), the overall amount is in the range of 55 to 57 electron-hole pairs per micron [8]. Increasing the device thickness and, therefore, the active volume would strengthen the signal. However, the achievable layer thicknesses are still constrained by the limitations in growing high-quality epitaxial layers. Consequently, sophisticated electronics is required to prepare the signals before post-processing.

Further constraints on the device thickness are given by the high intrinsic doping of approximately $10^{14}\,\mathrm{cm^{-3}}$ achieved in industry, which increases the full depletion and operational

This work has been supported by the Austrian Research Promotion Agency (FFG) in the project RadHardDetSim (895291).



(a) capacitance-voltage

(b) electric fields from anode (0 µm) to cathode (200 µm) at a reverse bias of 1 kV

Fig. 1. Simulations of a silicon diode for fluences from $3 \times 10^{14}\,\mathrm{n_{eq}/cm^2}$ to $1.3 \times 10^{16}\,\mathrm{n_{eq}/cm^2}$. Solid lines represent the simulation results from GTS and opaque lines those from Sentaurus. For a better fit, the fluences in the GTS simulations of the capacitance have been scaled with a factor of 0.3.

voltage compared to silicon. For currently available technologies approximately 500 V reverse bias is required to fully deplete a 50 um thick device. Even though 4H-SiC has a very high breakdown field strength, such high voltage values imply further design challenges such as the need for guard structures.

To characterize the behavior of particle detectors, three measurements are commonly conducted in high energy physics: I-V, C-V, and charge collection efficiency (CCE). Current measurements (I-V) in backward bias extract the dark current, which limits the detection sensitivity. In 4H-SiC the thermal charge carrier generation is, however, so minimal that the current is below 1 pA and limited experimentally by leakage paths and the resolution of the measurement device. The current in forward bias helps to classify the radiation damage, as described in the next section. Capacitance measurements (C-V) are used to evaluate the space charge region (active volume of the device), which can be used to extract the electrically active doping. Finally, the CCE is used to describe the ratio of a collected signal to the one achieved in a fully depleted and defect free device. Since the integral over the transient current induced by the moving charge carrier is equal to the deposited charge, the CCE denotes how efficient charge carriers can be collected.

To model the radiation damage in TCAD simulations, I-V, C-V, and CCE measurements of well known devices represent a crucial input. For this purpose we used the results reported by Gsponer *et al.* [1] who measured 4H-SiC detectors irradiated with 1 MeV equivalent neutron fluences between $10^{14}\,\mathrm{n_{eq}/cm^2}$ and $10^{16}\,\mathrm{n_{eq}/cm^2}$. The measurements revealed the surprising fact that the capacitance, which should change due to the varying width of the space charge region, stays constant in forward and reverse bias conditions (cp. Fig. 2a). In addition, the diodes can be heavily forward biased without showing significant conductance. While the onset of forward conductance in regular diodes starts around 2 V the radiated ones can even withstand more than 1 kV (cp. Fig. 3a).

The impact of radiation on 4H-SiC has been investigated by measurements in the past, however, the achieved results vary significantly. Overall, carbon vacancies, i.e., missing carbon atoms in the lattice, are the cause of the most prominent charge carrier traps $Z_{1,2}$ and $EH_{6,7}$. While the former is close to the conduction band [9], the latter is located in the middle of the bandgap [10], [11]. Besides the energy level, the type of the trap (donor or acceptor) and the cross sections for electrons and holes, i.e., the probability of a respective charge carrier entering the trap, are of importance. Unfortunately, the corresponding values in the literature still show a considerable spread, making a proper selection very challenging.

III. Simulations

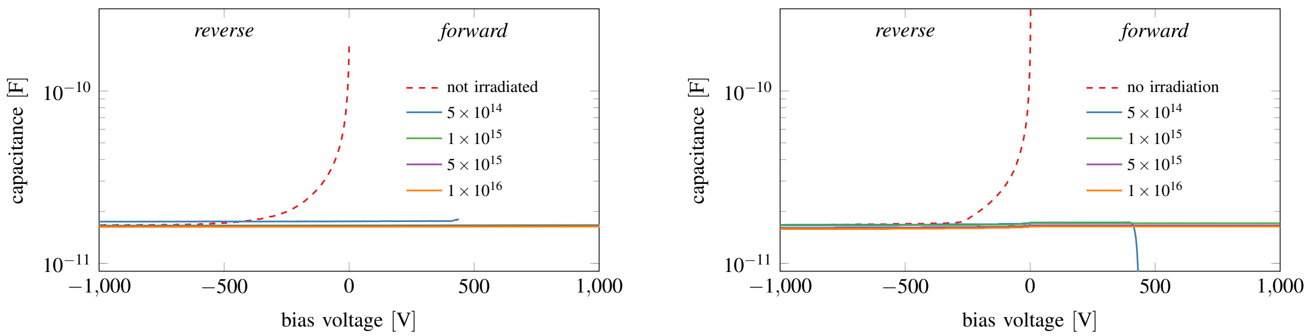
In this section, we present the TCAD simulation results, which have been achieved by using Synopsys® Sentaurus [12] in version 2023.09 and the simulation framework from Global TCAD Solutions (GTS) [6] in version 2024.03 Build 25. To verify the latter we first recreated the radiation damage simulations by Schwandt *et al.* [13] for silicon before turning 4H-SiC. Due to the tight page limit we will not explicitly show the achieved CCE results and just focus on I-V and C-V.

During our simulations with the GTS framework we encountered multiple convergence issues, which can be traced back to the still ongoing optimization of the tools. Consequently, we had to slightly adapt the simulation setup for each analysis. In the case of 4H-SiC, for example, incomplete ionization had to be disabled and the simulation grid was fine-tuned for forward and reverse bias separately.

A. Verification of Silicon

For the simulations of silicon, we used the Hamburg Penta Trap Model (HPTM) proposed in [13]. We want to highlight that we slightly simplified the structure of the investigated device, especially the doping profiles, so our results are not directly comparable to [13]. In detail a 200 µm thick 1D model of a pin-diode (bulk doping $3.0 \times 10^{12}\,\mathrm{cm^{-3}}$) with Gaussian-distributed doping profiles for the n+ ($N_D = 1.0 \times 10^{19}\,\mathrm{cm^{-3}}, \sigma = 1\,\mu\mathrm{m}$) and p+ ($N_A = 1.0 \times 10^{19}\,\mathrm{cm^{-3}}, \sigma = 1\,\mu\mathrm{m}$) contacts was used.

The simulation results of GTS fit qualitatively well to results achieved by simulations in Sentaurus. However, also



(a) measurements [1]

(b) GTS simulations

Fig. 2. Capacitance-voltage (C-V) characteristics of a 4H-SiC diode for fluences from $5 \times 10^{14}\,\mathrm{n_{eq}/cm^2}$ to $1 \times 10^{16}\,\mathrm{n_{eq}/cm^2}$. The rapid decrease in the simulated capacitance for a fluence of $5 \times 10^{14}\,\mathrm{n_{eq}/cm^2}$ near 500 V forward bias is due to the onset of conductance (cp. Fig. 3b).

differences, especially in regard to the radiation dose, were observed. For example, almost identical capacitance values have been achieved but for different fluences. In fact, scaling the latter by a factor of 0.3 led to a very good match (see Fig. 1a). In some cases GTS outperforms Sentaurus: For high reverse bias and neutron fluences above $3 \times 10^{15}\,\mathrm{n_{eq}/cm^2}$ the latter predicts an increase of the device's capacitance, which does not match the measurement results in [13]. Such an effect is not visible with GTS.

A key advantage of simulations compared to measurements is the ability to look inside the device. In Fig. 1b we show the simulated electric field across a reverse biased diode. Clearly visible is a double peak structure, whereat the field in the middle is almost zero. This can be explained by bulk traps that bind the majority charge carriers (electrons close to the cathode and holes close to the anode) and thus create a (quasi-stationary) space charge.

Although no perfect match among the tools was achieved, we conclude that the GTS framework is viable to provide qualitative trends in radiation damage. The quantitative differences can be, partially, explained by deviating implementations of the respective algorithms in each tool.

B. 4H-SiC Radiation Simulations

After verifying the silicon results, we turn to 4H-SiC. Respective measurements are shown in Fig. 2a and Fig. 3a. For the simulations we use the parameters in Table I, which represent a simplified version of the model used to investigate radiation damage with Sentaurus in [5].

The C-V and I-V simulation results are shown in Fig. 2b and Fig. 3b, respectively. For the former the match between measurements and simulations (simply scaled to the dimensions of the actual detector) is extraordinary. The decrease of the capacitance for increasing reverse bias vanishes completely. Instead, it stays at a constant low value, which only slightly increases at large current densities in forward bias. For a fluence of $5 \times 10^{14}\,\mathrm{n_{eq}/cm^2}$ the simulation fails to deliver reasonable results as the capacitance even becomes negative.

The I-V simulation results in Fig. 3b differ significantly from the measurements for reverse bias. This is a result of the very low leakage current that can be numerically calculated but is impossible to determine experimentally in the lab, as edge/surface effects and other leakage paths quickly dominate. Both results, however, agree in the fact that the irradiation has little impact on the dark current.

For forward bias, the decrease of conductance with radiation is clearly visible. The respective electric field shown in Fig. 4 highlights the differences to the reverse bias simulations for silicon discussed earlier. Although, again, field peaks at both sides of the pin-diode are visible, the field drops in between below zero. Our explanation is that injected charge carriers get trapped near their respective contact, forming effectively a space charge region. The resulting electric field hinders further electrons and holes from entering the device. At a sufficient forward bias all traps are filled, i.e., the field reaches a maximum, such that a further increase of the voltage results in a significant increases of the recombination current.

IV. Conclusion

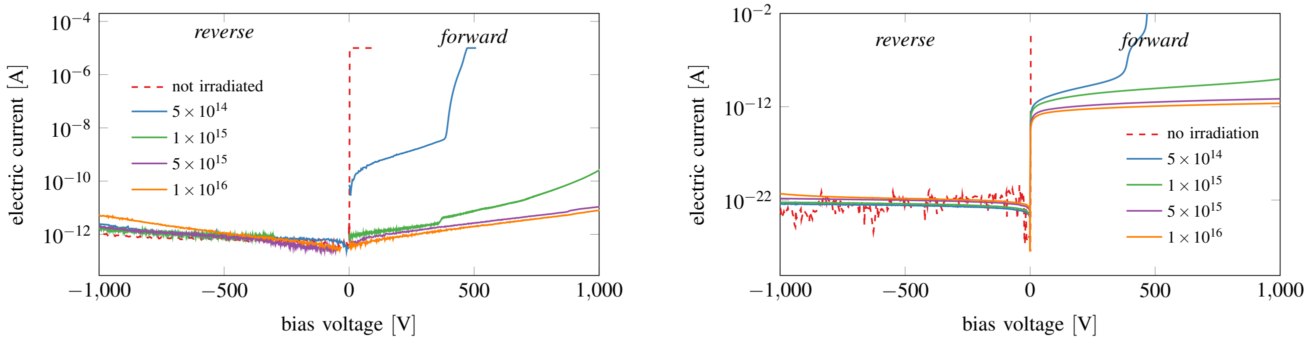
In this paper we presented a comparison of technology computer aided design (TCAD) simulations run with Sentaurus and the simulation framework from Global TCAD Solutions (GTS) against previously published simulation and measurement results of radiation damage in Si and SiC. Overall, the available capacitance and electric current data can be approximated closely. Based on the electric field deep inside the devices we were able to provide, for the first time, an explanation for the reduced conductance of radiated diodes in forward bias.

TABLE I

4H-SiC Radiation Damage Simulation Model [5]. $\sigma_{e,h}$ denote the electron/hole cross section, E_C the conduction band energy and g_{int} the introduction rate.

Defect	Type	Energy	g_{int} $[\mathrm{cm^{-1}}]$	σ_e $[\mathrm{cm^2}]$	σ_h $[\mathrm{cm^2}]$
$Z_{1,2}$	Acceptor	$E_C - 0.67\,\mathrm{eV}$ [a]	5.0 [b]	2e−14 [a]	3.5e−14 [a]
$EH_{6,7}$	Donor [c]	$E_C - 1.6\,\mathrm{eV}$ [d,e]	1.6 [b]	9e−12 [e]	3.8e−14 [d,e]
EH_4	Acceptor	$E_C - 1.03\,\mathrm{eV}$ [f,g]	2.4 [b]	5e−13 [g]	5.0e−14 [g]

[a] [9] [b] [14] [c] [15] [d] [10] [e] [11] [f] [16] [g] [17]



(a) measurements [1]

(b) GTS simulations

Fig. 3. Current-voltage (I-V) characteristics of a 4H-SiC diode for various fluences in n_{eq}/cm^2.

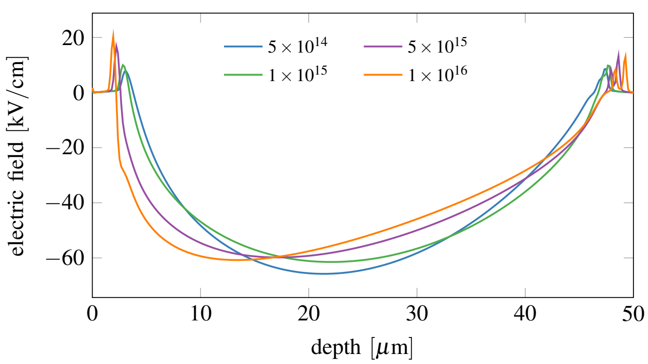


Fig. 4. Electric field along the 4H-SiC diode from anode (0 µm) to cathode (50 µm) in GTS for a forward bias of 200 V and various fluences in n_{eq}/cm^2. Explanation in the text.

In detail, trapped majority charge carriers effectively create a space charge region that hinders further charge carriers from entering the device.

In hindsight, our simulations show that the GTS framework is a suitable tool for radiation damage simulations in 4H-SiC. Our goal is to use such investigations in the development of next generation particle detector systems, i.e., to estimate the performance degradation during a life cycle. In cooperation with GTS we also plan to further improve the capabilities of the simulation framework, as occasionally nonphysical results have been achieved. Additional future research will be focused on a more thorough analysis of the introduced trap parameters, i.e., energy level, type, and cross-section, as literature values show large deviations.

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